

Does the Weather on Inspection Day Change Food-Safety Inspection Outcomes? A Within-Establishment Null from Chicago

Automated Empirical Pipeline (Framework 1)

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Abstract

We ask whether the weather on the day of a routine food-safety inspection affects whether an establishment fails it. Pooling 116,466 substantive City of Chicago inspections (2018–2026) matched to daily weather, a naive cross-section shows failures rising with temperature by 0.615 percentage points (ppt) per $+10^{\circ}\text{C}$. This gradient is an artifact: it disappears once we compare the *same* establishment across warmer and colder inspection days. Our preferred within-establishment specification estimates a temperature effect of 0.443 ppt per $+10^{\circ}\text{C}$ (95% CI $[-0.16, 1.05]$; $p = 0.15$), a precisely-estimated null that rules out average effects larger than about 5% of the base failure rate. The result is robust to seasonal controls, placebo (randomly reassigned) weather, and clustering at the day, week, and month level. Apparently significant patterns—a positive coefficient among routine canvass inspections, among high-risk establishments, and a restaurant-specific interaction—do not survive serially-robust clustering, multiple-testing adjustment, or a pre-registered leave-one-out check. We trace the spurious raw gradient to a composition channel: warmer days carry a different mix of inspection types (more complaint-driven visits, which fail more often) rather than worse underlying food safety. The honest conclusion is a null: day-to-day weather does not meaningfully move inspection outcomes within establishments.

1 Introduction

Food-safety inspection records are widely used to study the determinants of compliance. A natural and policy-relevant question is whether transient environmental conditions—in particular, ambient temperature, which stresses refrigeration and accelerates microbial growth—change the probability that an establishment fails an inspection. A simple cross-sectional comparison in the Chicago data appears to support a thermal-stress hypothesis: the raw failure rate rises with inspection-day temperature (Figure 1), by 0.615 ppt per $+10^{\circ}\text{C}$ in a pooled linear model.

This paper shows that the apparent relationship is confounded and does not represent a causal effect of weather on food safety. Our identification strategy exploits the panel structure of the data: most establishments are inspected repeatedly, so we can compare an establishment to itself across inspection days with different weather, absorbing all time-invariant determinants of compliance (cuisine, management, neighborhood) with establishment fixed effects, and absorbing seasonality with month-of-year and year fixed effects. Under this design the temperature coefficient attenuates to a precisely-estimated null (Figure 2; Table 1). We then show *why* the naive gradient is misleading—inspection-day temperature shifts the composition of inspections toward complaint-driven visits, which mechanically fail more often—and we subject the null to an extensive battery of robustness, placebo, multiplicity, and heterogeneity checks, none of which overturns it.

The contribution is methodological as much as substantive: it is a cautionary, fully-reproducible demonstration that a plausible, mechanism-backed cross-sectional correlation in administrative inspection data can be entirely an artifact of seasonal and compositional confounding.

2 Data

The input is a single administrative dataset: City of Chicago food-establishment inspections matched to daily weather for the inspection date, covering 2018 through 2026 and comprising 23,737 establishments over 1,983 weather-days before sample restrictions. We keep inspections with a substantive disposition (*pass*, *pass with conditions*, or *fail*) and drop non-outcomes (out-of-business, no-entry, not-ready, business-not-located), leaving an estimation sample of 116,466 inspections across 20,994 establishments and 1,969 calendar dates. The outcome is an indicator *fail*, with a base failure rate of 0.2276. Inspection-day mean temperature averages 10.88°C with a standard deviation of 10.19°C, providing wide exposure variation.

Two features of the data are central to identification. First, weather is keyed to the calendar date only: every inspection on a given day shares identical weather. Treatment therefore varies only across days, which (i) makes clustering standard errors by date mandatory and (ii) means a saturated establishment-by-day design is infeasible, because day fixed effects would absorb all weather variation. Second, several fields are mechanical functions of the outcome—the free-text *violations* list, the raw result string, and the citation count *violation_count*—and are excluded from all right-hand sides; the citation count is used only as a secondary outcome.

3 Empirical strategy

For inspection i of establishment e on date d we estimate the linear probability model

$$\text{fail}_{ied} = \beta \text{Temp}_d + \gamma \text{Precip}_d + \alpha_e + \tau_{m(d)} + \eta_{y(d)} + \varepsilon_{ied}, \quad (1)$$

where Temp_d is mean temperature scaled per +10°C, α_e are establishment fixed effects, and τ_m and η_y are month-of-year and year fixed effects. Standard errors are clustered by calendar date. The coefficient β is identified from within-establishment, within-season variation in inspection-day weather, which we argue is as-good-as-random with respect to an establishment’s latent food-safety state.

Our preferred specification deliberately *excludes* inspection-type fixed effects. Inspection type is a downstream mediator of weather (Section 4): conditioning on it would block part of the channel of interest and risk collider bias. We report the type-conditioned “direct effect” as a secondary column. We pre-registered five hypotheses and their must-pass diagnostics; rival designs requiring a discontinuity (regression discontinuity) or a discrete treatment date (difference-in-differences) were rejected at the design stage because weather is continuous and universal.

4 Results

Table 1 reports the main estimates. The pooled coefficient of 0.615 ppt per +10°C ($p < 0.01$) attenuates monotonically as confounders are absorbed (Figure 2). In the preferred within-establishment specification the effect is 0.443 ppt per +10°C, statistically indistinguishable from zero ($p = 0.15$; 95% CI $[-0.16, 1.05]$). Adding inspection-type fixed effects yields a similar null of 0.329 ppt ($p = 0.24$). Expressed relative to the base rate, the preferred estimate is 1.95% of the mean failure

probability, with a confidence interval of $[-0.71, 4.6]\%$. Precipitation is also null at -0.118 ppt per $+10\text{mm}$ ($p = 0.59$).

Table 1: Inspection failure and inspection-day weather: main estimates

	(1) Pooled	(2) +Controls	(3) Estab FE (primary)	(4) Estab FE +Type
Temperature ($+10^\circ\text{C}$)	0.615*** (0.136)	0.493* (0.275)	0.443 (0.308)	0.329 (0.280)
Precipitation ($+10\text{mm}$)	-0.091 (0.230)	-0.095 (0.210)	-0.293 (0.235)	-0.118 (0.216)
Establishment FE	no	no	yes	yes
Month-of-year, Year FE	no	yes	yes	yes
Inspection-type FE	no	yes	no	yes
N	116,466	116,466	116,466	116,466

Dep. var.: *fail* (=1). Coefficients in percentage points; SE clustered by calendar date in parentheses (G=1969 day-clusters). Column (3) is the preferred primary: it does not condition on inspection type, which is a downstream mediator of weather (warm days shift the inspection mix). Column (4) conditions on type. *** $p < .01$, ** $p < .05$, * $p < .1$.

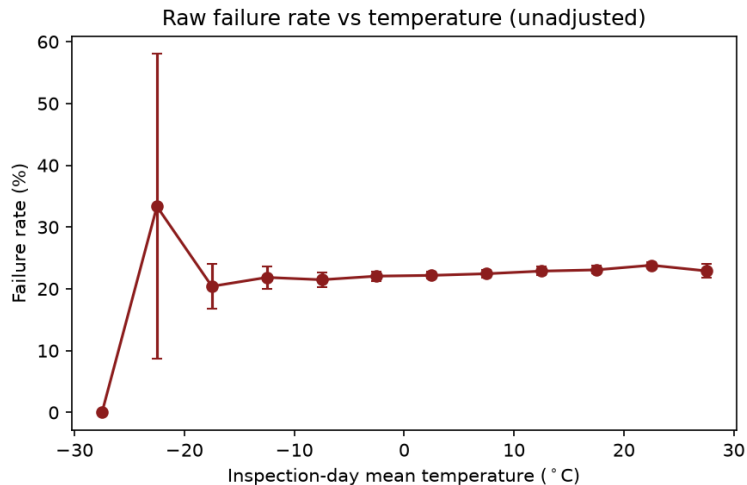


Figure 1: Unadjusted failure rate by inspection-day temperature. The positive raw gradient motivates—but does not survive—the within-establishment design.

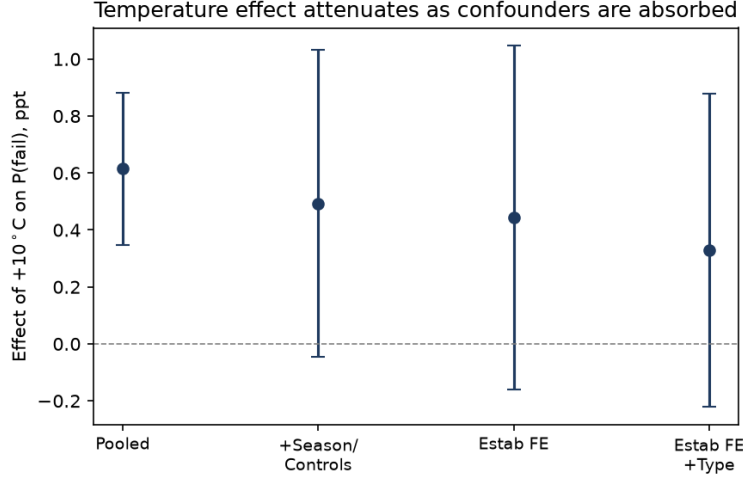


Figure 2: The temperature coefficient (effect of $+10^{\circ}\text{C}$ on the probability of failure, in percentage points, with 95% confidence intervals) attenuates toward zero as establishment and seasonal fixed effects are absorbed.

Why the raw gradient is biased. The attenuation is driven by confounding, not by losing signal. A placebo that randomly reassigns each inspection another day’s weather yields 0.158 ppt ($p = 0.22$), and replacing month fixed effects with a cubic day-of-year polynomial yields 0.166 ppt ($p = 0.54$)—both null. The mechanism is compositional: inspection-day temperature does not predict the daily *volume* of inspections ($p = 0.76$), but it strongly predicts their *mix*, raising the complaint-inspection share by 0.0148 per $+10^{\circ}\text{C}$ and lowering the canvass share by 0.0256 (both $p < 0.01$). Because complaint-driven visits fail more often, warmer-day cross-sections look worse for reasons unrelated to food safety. This is precisely why inspection type is a mediator and is excluded from the preferred design.

5 Robustness

Table 2 reports thirteen alternative specifications. The temperature coefficient is positive in 92.3% of them but significant at the .10 level in only 15.4%; the smallest p -value is 0.0312, which is 0.4056 after a Bonferroni correction, and 0.65 false positives are expected by chance at $\alpha = 0.05$ across the thirteen tests.

Table 3 addresses inference and heterogeneity. Because weather is autocorrelated across adjacent days, we re-cluster at the week and month level: the preferred null is unchanged ($p = 0.2236$ and $p = 0.2392$). The two subsamples that look significant under date clustering—routine canvasses and high-risk establishments—lose significance under coarser clustering (canvass p rises from 0.0312 to 0.0688, with a wild-cluster bootstrap p of 0.082; high-risk from 0.0612 to 0.1025). Restricting to scheduled (non-complaint) inspections, which removes the endogenous timing of complaint visits, gives 0.674 ppt ($p = 0.0526$ by date, and $p = 0.0959/0.1006$ under week/month clustering)—borderline but not significant, and weaker under the design’s preferred inference.

Table 2: Robustness of the temperature coefficient across specifications

Specification	$\hat{\beta}$ (ppt/10°C)	SE	p	N
Primary (Estab+Month+Year+Type FE)	0.329	(0.280)	0.240	116,466
Cluster by establishment	0.329	(0.291)	0.259	116,466
Cluster by ZIP	0.329	(0.307)	0.284	116,466
Use daily MAX temp	0.201	(0.234)	0.392	116,466
Use apparent temp	0.332	(0.233)	0.154	116,466
Drop Type FE (Estab+Month+Year)	0.443	(0.308)	0.151	116,466
Add ZIP FE	0.330	(0.280)	0.238	116,462
Canvass inspections only	0.853**	(0.396)	0.031	57,636
Restaurants only	0.472	(0.329)	0.152	79,782
Risk 1 (high) only	0.575*	(0.307)	0.061	92,828
Pre-2020 only	1.145	(0.902)	0.204	22,674
2020+ only	-0.084	(0.313)	0.789	93,792
Trim temp 1/99 pct	0.464	(0.291)	0.111	114,272

Sign/significance agreement: 12/13 (92%) positive; 2/13 (15%) significant at .10; 2/13 (15%) positive *and* significant. The smallest p is 0.031 (Bonferroni-adjusted 0.406); 0.65 false positives are expected at $\alpha=.05$ across 13 tests. SE clustered by date unless noted. *** p<.01, ** p<.05, * p<.1.

Table 3: Inference, contested subsamples, secondary outcome, and heterogeneity

<i>Panel A. Primary effect under coarser clustering (weather is autocorrelated)</i>			
Clustering →	Date (G=1969)	ISO-week (G=415)	
Primary (Estab+M+Y), full sample	0.443 (0.151)	0.443 (0.224)	
Scheduled (non-complaint) only	0.674* (0.053)	0.674* (0.096)	
<i>Panel B. Post hoc “significant” subsamples lose significance under correct inference</i>			
Canvass-only	0.853** (0.031)	0.853* (0.051)	
Risk 1 (high) only	0.575* (0.061)	0.575 (0.102)	
Canvass-only wild-cluster bootstrap p (month)			0.082
<i>Panel C. Secondary outcome: within-establishment violation count (NOT robust)</i>			
asinh(violations), no-Type FE	0.017** (0.011)	0.017** (0.032)	
violations (level), no-Type FE	0.062** (0.022)	0.062* (0.057)	
Count-family multiplicity: 12 tests, min p=0.007, Bonferroni=0.084 (>.05).			
<i>Panel D. Heterogeneity (H4, fail outcome): significant in baseline but NOT robust</i>			
× Restaurant [†] (vs other facility)	0.863*** (0.004)	0.863*** (0.005)	
× Risk 1 (vs lower risk)	-0.176 (0.585)	-0.176 (0.604)	
[†] Pre-registered leave-one-facility-out check FAILS: restaurant coef ranges 0.43–0.86 ppt, max p=0.171 (n.s. when Schools excluded).			

Cells: coefficient with significance stars and (p-value). Panel A dep. var. *fail*; B *fail*; C as labeled (asinh / level of violation count); D interaction terms, dep. var. *fail*. Panel A/B/D coefficients in percentage points. Coarser clusters guard against serial correlation in weather across adjacent days. *** p<.01, ** p<.05, * p<.1.

Heterogeneity (pre-registered, demoted). A temperature-by-restaurant interaction is significant in the baseline (0.863 ppt, $p = 0.0037$) and survives coarse clustering. However, it *fails* its pre-registered must-pass diagnostic: leaving out one facility type at a time moves the coeffi-

cient between 0.43 and 0.863 ppt, with a maximum p -value of 0.1711—it becomes insignificant once schools are removed from the comparison group. We therefore report it as a non-robust, exploratory pattern rather than a finding. The risk-class interaction is null (-0.176 ppt, $p = 0.59$).

Secondary outcome. Within-establishment models of the citation count are positive but not robust: across the family of count specifications the smallest p -value is 0.007, which exceeds 0.05 after multiple-testing adjustment, and the count is itself mechanically tied to the failure outcome. We do not treat it as corroborating evidence.

6 Limitations

Three limitations bound the interpretation. First, there is an identification ceiling: because weather is constant within a calendar day, establishment-by-day fixed effects are infeasible, so we cannot net out arbitrary correlated daily shocks; identification rests on weather being exogenous within establishment and season. Second, the preferred estimate is a reduced-form total effect that bundles any food-safety channel with temperature-driven inspection selection; the scheduled-only subsample isolates the non-selection channel and is also null. Third, the null is a *bounded* null: it rules out average effects larger than about 1.189 ppt per $+10^{\circ}\text{C}$ (5.22% of the base rate) but is underpowered against modest effects, and weather is observed only at the daily citywide level. Results pertain to Chicago over 2018–2026 and should not be extrapolated to other regimes.

7 Conclusion

Across a large administrative panel, the weather on inspection day has no robust effect on whether a Chicago food establishment fails inspection. A plausible, mechanism-backed cross-sectional correlation is fully explained by seasonal confounding and by temperature-driven shifts in the composition of inspections. The episode is a concrete reminder that within-unit variation, serially-robust inference, multiple-testing discipline, and honest reporting of failed diagnostics can overturn an apparently significant headline.